

Supporting Information

Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting,
Environmental Calibration, and Agro-Rural Cybersecurity

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S1 8-Site FMO Hamiltonian and full coupling matrix

S1.1 Hamiltonian formulation

The 8-site Fenna–Matthews–Olson (FMO) Hamiltonian used throughout this work extends the standard 7-site model [1] by including the eighth bacteriochlorophyll *a* molecule that bridges the chlorosome baseplate to the reaction center [2]. The full electronic Hamiltonian in the site basis is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (\text{S1})$$

where ε_n are the site energies (referenced to the lowest-energy site), J_{nm} are the inter-site electronic couplings, and $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$ is the irreversible reaction center trapping rate applied to Sites 3 and 4. The anti-Hermitian term $-i\Gamma_{\text{RC}}$ models irreversible charge separation at the reaction center; the Hamiltonian is non-Hermitian only on these diagonal trapping terms.

S1.2 Site energies

Table S1 lists the site energies used in all production simulations. Energies are given relative to the lowest-energy site (Site 3 at $12\,210 \text{ cm}^{-1}$) and are expressed in cm^{-1} .

Table S1: **8-site FMO site energies** (in cm^{-1} , referenced to Site 3).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
Absolute ε_n	12 410	12 530	12 210	12 320	12 480	12 630	12 440	12 410
Relative ε_n	280	420	0	110	270	500	310	200

The absolute site energies are taken from the Adolphs–Renger parameterization [1] for Sites 1–7. Site 8, which bridges the baseplate to the reaction center, is assigned an energy of $12\,410 \text{ cm}^{-1}$ based on the eighth-bacteriochlorophyll characterization by Moix *et al.* [2].

S1.3 Electronic coupling matrix

Table S2 presents the full 8×8 electronic coupling matrix J_{nm} (in cm^{-1}). The matrix is symmetric ($J_{nm} = J_{mn}$), and all 28 unique coupling pairs are listed. Values for Sites 1–7 follow the Adolphs–Renger parameterization [1]; couplings involving Site 8 are taken from Ref. [2].

Table S2: **Full 8-site electronic coupling matrix** J_{nm} (in cm^{-1}).

	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
Site 1	0	−87.7	5.5	−5.9	6.7	−13.7	−9.9	4.0
Site 2		0	30.8	8.2	0.7	11.8	4.3	−2.1
Site 3			0	−53.5	−2.2	−9.6	6.0	12.0
Site 4				0	−70.7	−17.0	−63.3	5.5
Site 5					0	81.1	−1.3	3.2
Site 6						0	39.7	−8.7
Site 7							0	−15.0
Site 8								0

S1.4 Reaction center trapping

The irreversible trapping is implemented through a non-Hermitian term on Sites 3 and 4, which together form the primary funnel to the reaction center in the 8-site FMO model [3, 2]. The trapping rate $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$ corresponds to a trapping time of $\tau_{\text{RC}} \approx 6.7 \text{ ps}$, consistent with ultrafast charge separation rates measured in reaction center complexes [3]. The forward transfer yield is defined as:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} [P_3(t) + P_4(t)] dt, \quad (\text{S2})$$

with $t_{\text{max}} = 1 \text{ ps}$ and $P_j(t) = \rho_{jj}(t)$. Sites 3 and 4 are the two trapping sites of the network, and both are summed in the integrand.

S1.5 Mapping FMO model dynamics to higher plant LHCII complexes

The 8-site FMO complex of green sulfur bacteria serves as our primary model system due to its well-characterized excitonic parameters, which make it a standard benchmark in quantum biology. To establish the generality of our agrivoltaic framework for crops such as strawberries and cut flowers, the physical principles of vibronic resonance-matching are mapped directly onto the light-harvesting complex II (LHCII) of higher plants.

The primary light-harvesting complex in crop leaves, LHCII, consists of Chlorophyll *a* (Chl *a*) and Chlorophyll *b* (Chl *b*) chromophores embedded in a protein matrix. These chromophores exhibit strong coupling to underdamped intra-molecular vibrational modes. Specifically, the ring-breathing and macrocycle modes of chlorophylls generate prominent underdamped vibronic peaks around $180\text{--}210 \text{ cm}^{-1}$ and $1100\text{--}1200 \text{ cm}^{-1}$, mirroring the Kleinekathöfer/Coker 12-mode bath model of bacteriochlorophyll *a* in FMO.

Table S3 compares the key physical parameters of FMO to those of crop LHCII. While the absolute exciton transition wavelengths shift from the near-infrared (800 nm) to the visible red (650–670 nm), the relative energy scales of electronic coupling (J_{nm}), vibronic frequencies (ω_{vib}), and bath reorganization energies (λ_D) are highly conserved.

Table S3: **Biophysical parameter comparison between bacterial FMO and higher plant LHCII complexes.**

Physical Parameter	FMO (BChl <i>a</i>)	LHCII (Chl <i>a</i>)	LHCII (Chl <i>b</i>)
Primary Absorption Peak	800 nm	670 nm	650 nm
Key Vibronic Mode Frequency ω_{vib}	180 cm^{-1}	195 cm^{-1}	210 cm^{-1}
Drude-Lorentz Reorganization λ_D	35 cm^{-1}	30 cm^{-1}	25 cm^{-1}
Inter-site Coupling Range J_{nm}	$-100\text{--}100 \text{ cm}^{-1}$	$-120\text{--}120 \text{ cm}^{-1}$	$-90\text{--}90 \text{ cm}^{-1}$
Vibronic Coupling Strength g_{vib}	0.02 to 0.06	0.03 to 0.05	0.02 to 0.04

Under the resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$, the dual-band spectral filter for crops must be shifted from 750–820 nm to 630–680 nm to match the absorption profile of crop chlorophylls. The active OPV organic materials are selected to exhibit a narrow transmission passband in the 630–680 nm window. The coherence protection mechanism remains mathematically identical.

S1.6 Hermiticity validation

The non-Hermiticity is strictly confined to the diagonal entries of Sites 3 and 4. For all $i \neq j$, $H_{ij} = H_{ji}^*$ is satisfied within machine precision (1×10^{-14}). The Hamiltonian’s Hermiticity condition is verified for all 64 matrix elements at every time step of the simulation.

S2 Convergence analysis

S2.1 Hierarchy depth convergence

We adopt $L = 8$ as the production standard to ensure robust convergence across all disorder realizations. The numerical framework (PT-HOPS/SBD implementation, hierarchy equations) is directly inherited from the validation framework established in Paper I [4], where convergence was validated via MAE analysis on the same excitonic parameters.

Table S4 summarizes the mean absolute error (MAE) between successive L values and the relative deviation from the $L = 9$ reference (computed for a subset of 10 disorder realizations).

Table S4: **Hierarchy depth convergence.** MAE between successive hierarchy depths for exciton populations, averaged over all 8 sites and 100 disorder realizations.

L	MAE ($L \rightarrow L + 1$)	Rel. dev. from $L = 9$	CPU time (min)
6	9.8×10^{-10}	1.2×10^{-8}	2.1
7	9.6×10^{-10}	1.1×10^{-8}	4.3
8	3.1×10^{-11}	4.2×10^{-10}	7.5

The MAE between $L = 7$ and $L = 8$ is 3.1×10^{-11} , which is below double-precision round-off for most observables. We adopt $L = 8$ as the production standard to ensure robust convergence across all disorder realizations.

S2.2 Matsubara truncation

We test Matsubara term counts $K = 1, 2, 3$ to assess the impact of high-temperature corrections. Table S5 shows the effect on the coherence lifetime τ_c and the forward transfer yield Φ_{FT} .

Table S5: **Matsubara truncation convergence.** Impact of K on key observables at 295 K.

K	τ_c (fs)		Φ_{FT}	MAE from $K = 3$
1	406	± 30	0.86 ± 0.03	3.4×10^{-4}
2	420	± 35	0.89 ± 0.03	3.3×10^{-5}
3	422	± 36	0.89 ± 0.03	—

The MAE between $K = 2$ and $K = 3$ is 3.3×10^{-5} , confirming that $K = 2$ is sufficient for converged dynamics at room temperature. This is consistent with the expectation that at 295 K, the dynamics are dominated by the explicit vibronic modes rather than high-order Matsubara corrections to the Drude–Lorentz background.

S2.3 Time step convergence

We verify time step convergence by comparing $\Delta t = 0.2$ fs, 1.0 fs and 2.0 fs. Table S6 summarizes the results.

Table S6: **Time step convergence.** Maximum population deviation relative to the $\Delta t = 0.2$ fs reference.

Δt (fs)	Max. population deviation
0.2	— (reference)
1.0	4.3×10^{-3}
2.0	8.1×10^{-3}

We adopt $\Delta t = 0.2$ fs as the production standard, consistent with Paper I [4].

S2.4 Detailed balance

The steady-state populations are compared with the analytical Boltzmann distribution to verify detailed balance. The maximum deviation across all 8 sites is 2.3 %, confirming that thermal equilibrium is correctly reproduced.

S2.5 Physical consistency checks

All production simulations satisfy the following consistency checks:

- **Trace preservation:** $\max_t |\text{Tr}[\rho(t)] - 1| < 1 \times 10^{-12}$
- **Positivity:** $\min_i \lambda_i(t) > -1 \times 10^{-14}$ for all eigenvalues λ_i of $\rho(t)$
- **Hermiticity:** $\max_{i,j,t} |\rho_{ij}(t) - \rho_{ji}^*(t)| < 1 \times 10^{-14}$
- **Population realness:** $\max_{i,t} |\text{Im}[\rho_{ii}(t)]| < 1 \times 10^{-14}$

S3 Polaron dressing mechanism and residual reorganization energy

S3.1 Canonical (lang–firsov) transformation

The polaron dressing mechanism that underlies the coherence protection reported in the main text can be rigorously understood through the canonical Lang–Firsov transformation [5, 6]. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k} (\hat{b}_{n,k}^\dagger + \hat{b}_{n,k}) + \hat{H}_{\text{bath}}, \quad (\text{S3})$$

where $g_{n,k}$ are the system-bath coupling constants for mode k at site n , and $\hat{b}_{n,k}$ ($\hat{b}_{n,k}^\dagger$) are the bath annihilation (creation) operators.

The unitary transformation $\hat{U} = \exp\left[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k} (\hat{b}_{n,k}^\dagger - \hat{b}_{n,k})\right]$ partially diagonalizes the system-bath coupling. After applying $\hat{U}$, the transformed Hamiltonian becomes:

$$\hat{H}' = \hat{U}^\dagger \hat{H} \hat{U} = \hat{H}'_S + \hat{H}_{\text{bath}} + \hat{H}_{\text{res}}, \quad (\text{S4})$$

where $\hat{H}'_S$ contains renormalized excitonic couplings (the polaron-shifted energies) and $\hat{H}_{\text{res}}$ represents the residual system-bath coupling after the transformation.

S3.2 Residual spectral density

The key physical insight is that underdamped vibronic modes satisfying the resonance condition $\Delta E_{nm} \approx \hbar\omega_v \pm J_{nm}$ become partially “dressed” by the canonical transformation: they are incorporated into $\hat{H}'_S$ and no longer contribute to dephasing in the transformed frame. The residual spectral density is:

$$J_{\text{res}}(\omega) = J_{\text{bath}}(\omega) - \sum_{k \in \mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}, \quad (\text{S5})$$

where $\mathcal{S}$ denotes the set of underdamped modes that are vibronically dressed.

S3.3 Quantitative effect

For the 12-mode Kleinekathöfer/Coker parameterization at 295 K, the residual reorganization energy under the two excitation scenarios is:

$$\lambda_{\text{res}}^{\text{broad}} = \lambda_{\text{DL}} + \sum_{k=1}^{12} S_k \omega_k \approx 53 \text{ cm}^{-1}, \quad (\text{S6})$$

$$\lambda_{\text{res}}^{\text{filt}} = \lambda_{\text{DL}} + \sum_{k \notin \mathcal{S}} S_k \omega_k \approx 40 \text{ cm}^{-1}. \quad (\text{S7})$$

The 20 % reduction in λ_{res} translates to a 50 % coherence lifetime extension (since $\tau_c \propto 1/\lambda_{\text{res}}$ in the high-temperature limit). The dual-band filter at 750 nm and 820 nm selectively excites those vibronic resonances that are most strongly coupled to the energy-transfer pathway (primarily the 180 cm^{-1} Franck–Condon mode at Sites 3/4 and the 740 cm^{-1} ring deformation mode), maximizing the dressing effect.

S3.4 Connection to PT-HOPS/SBD

In the PT-HOPS/SBD simulation framework, this protection is captured natively: the bath correlation function is decomposed into Drude–Lorentz and underdamped components, and the hierarchy equations propagate the system-bath entanglement without assuming weak coupling. The resulting trajectories exhibit the suppressed decoherence predicted by the vibronic-basis analysis when the pulse spectrum matches the underdamped resonances.

S4 Dynamic photo-protection: Floquet–Stark and OMIT behaviour

S4.1 Floquet–Stark detuning

Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant changes dynamically, inducing a time-dependent Stark shift. The driving term in the Hamiltonian is:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_F t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (\text{S8})$$

with $V_0 = 0.05 \text{ eV}$ and $\omega_F = 1.2 \text{ THz}$. At $t = 0$ (zero flux), the cosine term evaluates to V_0 , ensuring continuity with the undriven Hamiltonian. When the flux drops below threshold, the amplitude V_0 decays to zero with a relaxation time $\tau_{\text{relax}} = 10 \text{ ps}$, preventing numerical singularities.

S4.2 OMIT transmission modulation

The optomechanically induced transparency (OMIT) mechanism [7] creates destructive interference in the NPoM cavity that suppresses transmission at the resonant channels. The transmission coefficient is:

$$T_{\text{OMIT}}(I) = \begin{cases} T_0, & I = 0, \\ T_0 \left(1 + \frac{I}{I_{\text{sat}}}\right)^{-1}, & I > 0, \end{cases} \quad (\text{S9})$$

where $I_{\text{sat}} = 800 \text{ W m}^{-2}$. The piecewise definition at $I = 0$ ensures continuity: $T_{\text{OMIT}}(0) = T_0$, corresponding to the undriven cavity transmission. At high flux (1000 W m^{-2}), $T = T_0 \times (1 + 1000/800)^{-1} = 0.44 T_0$, i.e., transmission drops to 44 % of baseline. The time-dependent flux follows 24 h solar cycles, and the OMIT response is instantaneous on agricultural timescales.

We emphasize that the OMIT-induced transmission reduction acts by coherent back-reflection (increasing the local albedo of the NPoM sentinel nodes) rather than inelastic absorption. This prevents localized thermal heating in the sub-nanometer gap, which would otherwise degrade the zwitterionic polymer coating and denature the biological complexes.

S4.3 Night mode (zero flux)

At zero flux (night mode), both photo-protection mechanisms return to their baseline states:

- Floquet Stark: $V_0 \rightarrow 0$, restoring the undriven OPV–FMO coupling.
- OMIT: $T \rightarrow T_0$, restoring full transmission.

No singularities or discontinuities arise because both models are explicitly defined for $I = 0$.

S5 Socioeconomic analysis: cooperative payback model

S5.1 Capital expenditure and cost structure

The capital expenditure (CAPEX) for the quantum agrivoltaic system (Scenario A) comprises:

- **NPoM-coated graphene barrier:** 120 m^{-2} (including gold nanoparticle synthesis and deposition)
- **OPV module:** 80 m^{-2} (blade-coated semi-transparent polymer cells)
- **Buried single-mode fiber-optic link:** 30 m^{-2} (trenching, fiber deployment, and termination)
- **Installation and mounting:** 35 m^{-2} (optimized structures)
- **Electronics and IoT:** 25 m^{-2} (sensors, microcontroller, QKD transceiver module)
- **Contingency (10 %):** 32 m^{-2}

Total CAPEX: 322 m^{-2} .

S5.2 Cooperative ownership model

We model a cooperative of N smallholder farmers who collectively own and operate the quantum agrivoltaic system. The system is scaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers), yielding a total CAPEX of 161 000 USD. With a blended-finance subsidy s , the net CAPEX is $C_{\text{net}} = C_{\text{total}}(1 - s)$.

The payback period is computed as:

$$P = \frac{C_{\text{total}}(1 - s)}{R_{\text{total}} - C_{\text{OPEX}}}, \quad (\text{S10})$$

where C_{OPEX} comprises extension services and technician training ($1\,200 \text{ USD yr}^{-1}$), OPV panel cleaning to mitigate soiling (800 USD yr^{-1}), and general maintenance ($2\,000 \text{ USD yr}^{-1}$), totaling $4\,000 \text{ USD yr}^{-1}$.

S5.3 Revenue streams

- **Electricity:** $180 \text{ kWh m}^{-2} \text{ yr}^{-1}$ at $0.12 \text{ kWh}^{-1} = 21.6 \text{ m}^{-2} \text{ yr}^{-1}$
- **Water savings:** $1300 \text{ L m}^{-2} \text{ yr}^{-1}$ at $0.002 \text{ L}^{-1} = 2.6 \text{ m}^{-2} \text{ yr}^{-1}$
- **Crop yield (high-value floriculture):** $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$ at $4.5 \text{ USD kg}^{-1} = 36.9 \text{ m}^{-2} \text{ yr}^{-1}$. The 15 % quantum improvement contributes $5.5 \text{ m}^{-2} \text{ yr}^{-1}$.
- **Carbon credit (conservative):** $19.6 \text{ kgCO}_2\text{e m}^{-2} \text{ yr}^{-1}$ at $10 \text{ USD t}^{-1} = 0.196 \text{ m}^{-2} \text{ yr}^{-1}$

For a 500 m^2 module, the total annual revenue is approximately 30 612 USD. Subtracting the 4000 USD OPEX yields a net annual cash flow of 26 612 USD.

S5.4 Payback periods and npv

For a 500 m^2 module under a 30 % blended-finance subsidy ($C_{\text{net}} = 112\,700 \text{ USD}$), the simple payback period is $112\,700 \text{ USD} / 26\,612 \text{ USD yr}^{-1} \approx 4.23 \text{ yr}$ for the single-owner baseline (or 6.05 yr without subsidy). Under the cooperative risk-sharing model, the 5 members pool resources and share the same subsidy, so each member’s individual capital exposure drops to 22 540 USD at an identical 4.23 yr payback period — the cooperative’s primary benefit is risk diversification and a lower barrier to entry, not a shorter payback per se. The 10-year discounted NPV at a 12 % risk-adjusted discount rate appropriate for Sub-Saharan Africa [8] is 37 664 USD when carbon credits are included. Although a baseline CAPEX of 322 m^{-2} is high relative to Sub-Saharan African living standards, localized adaptation — such as using bamboo/timber supports instead of steel and green synthesis of gold nanoparticles from local plant extracts — can reduce capital costs to 140 m^{-2} , reducing the unsubsidized payback period to 2.63 yr.

We propose a tiered subsidy framework where the NEB serves as the verifiable metric for subsidy qualification:

- **Tier 1 (NEB < 5 kgCO₂e m⁻² yr⁻¹):** 0 % subsidy
- **Tier 2 (NEB 5–10 kgCO₂e m⁻² yr⁻¹):** 15 % subsidy
- **Tier 3 (NEB > 10 kgCO₂e m⁻² yr⁻¹):** 30 % subsidy

This creates a direct incentive loop: higher quantum yield improvements lead to higher NEB, which unlocks greater subsidy access, enabling broader deployment. To prevent capital loss and ensure technological accountability, Tier 3 subsidies must be structured as Performance Bonds or incorporate sunset provisions. Rather than providing unconditional upfront capital grants, the development bank loans are amortized or partially forgiven post-facto only if the co-deployed QKD and SERS sentinel network certifies (via the permissioned blockchain ledger) that the targeted water savings and crop yield thresholds have been sustained for at least 24 months.

S6 BB84 quantum key distribution: implementation and performance

S6.1 Protocol details

The BB84 protocol [9] is implemented as follows:

1. **Preparation:** Alice (sensor node) generates $4N = 1024$ random bits (for $N = 256$ final key length) and encodes each bit in one of four polarization states: $|H\rangle$, $|V\rangle$ (rectilinear basis, +) or $|D\rangle$, $|A\rangle$ (diagonal basis, ×).

2. **Transmission:** The quantum states are transmitted over a buried single-mode fiber channel with a simulated depolarizing noise rate of $p = 4.8\%$, modeling the baseline error rate in field-deployed fiber links [10].
3. **Measurement:** Bob (cloud server) measures each received qubit in a randomly chosen basis (+ or $\times$), independent of Alice’s preparation basis.
4. **Sifting:** Alice and Bob publicly announce their basis choices and retain only events where bases match (approximately $2N = 512$ bits remain).
5. **Parameter estimation:** A random sample (minimum 1 bit, maximum 50 bits) of the sifted key is disclosed to estimate the quantum bit error rate (QBER). The sample fraction follows $f_{\text{sample}} = \min(50/N, 0.2)$.
6. **Key generation:** If $\text{QBER} < 11\%$ (the Shor–Preskill security threshold), the remaining sifted bits form the raw key. Error correction and privacy amplification reduce the key to $N = 256$ bits. We specify that to secure a final 256 bit key under a 4.8 % QBER, Alice prepares a block of $N_{\text{pulse}} \approx 2200$ pulses. Sifting discards 50 % of the bits due to basis mismatch, leaving $N_{\text{sift}} \approx 1100$. The secure key rate is bounded by $r = 1 - h_2(e) - f(e)h_2(e) \approx 0.397$, where $h_2(e) = 0.274$ is the binary entropy and $f(e) \approx 1.2$ is the error correction efficiency of Cascade/LDPC reconciliation. This yields $N_{\text{secure}} \approx 436$ bits, which converges to a 256 bit key after accounting for finite-size effects and disclosure during parameter estimation.

S6.2 Error threshold and security

The 11 % QBER threshold is derived from the intercept-resend attack bound: an eavesdropper (Eve) performing intercept-resend on all qubits introduces a minimum QBER of 25 %. The 11 % threshold provides unconditional security via the Shor–Preskill proof [11].

S6.3 Fail-Safe mechanism

If the QBER exceeds 11 % (e.g., due to equipment malfunction or attempted eavesdropping on the fiber link), the key generation aborts via a `SecurityThresholdExceeded` exception, and a fail-safe local irrigation routine activates:

- QKD-protected cloud irrigation commands are blocked to prevent data-injection attacks.
- A local fallback executes a conservative irrigation schedule based on the last-validated soil moisture reading from the *DynamicCalibrator*.
- Telemetry data is cached locally and transmitted when the secure channel is restored.

S6.4 Computational overhead

Table S7 shows the runtime overhead of the BB84 simulation for a resource-limited IoT micro-controller (simulated on a single ARM Cortex-M4 core at 120 MHz).

The total runtime of 4.8 ms and memory footprint of 2.2 kB are negligible for resource-limited IoT nodes, confirming the practical feasibility of BB84-secured agricultural telemetry.

Table S7: **BB84 QKD simulation runtime overhead.**

Operation	Time (ms)	Memory (kB)
State preparation (1024 bits)	0.8	0.5
Noisy transmission	1.2	0.1
Measurement and sifting	0.6	0.3
QBER estimation	0.1	0.1
Key distillation	2.1	1.2
Total	4.8	2.2

S7 Generalized resonance condition for vibronic coherence protection

S7.1 Theoretical derivation

The resonance condition that enables the coherence protection mechanism is a general property of vibronically coupled chromophore aggregates [5, 12]. For a dimer of two chromophores with site energies ε_1 , ε_2 and electronic coupling J_{12} , the excitonic energy gap is:

$$\Delta E_{\text{exc}} = \sqrt{(\varepsilon_1 - \varepsilon_2)^2 + 4J_{12}^2}. \quad (\text{S11})$$

When this gap is resonant with an underdamped vibrational mode of frequency ω_v :

$$\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{12}, \quad (\text{S12})$$

the excitonic and vibrational states hybridize to form a vibronic (polaron-dressed) state. The $\pm J_{12}$ term accounts for Davydov splitting in the dimer: the symmetric and antisymmetric combinations of the two site excitations have energies separated by $2J_{12}$, each coupling differently to the vibrational manifold.

In the vibronic basis (after the Lang–Firsov transformation), the effective pure-dephasing rate between excitonic states $|\alpha\rangle$ and $|\beta\rangle$ is:

$$\Gamma_{\alpha\beta}^{\text{pd}} = \frac{1}{\hbar^2} \int_0^\infty d\omega \frac{J_{\text{res}}(\omega)}{\omega^2} \left[\coth\left(\frac{\beta\hbar\omega}{2}\right) (1 - \cos\omega t) + i \sin\omega t \right] \Big|_{t \rightarrow \infty}, \quad (\text{S13})$$

where $J_{\text{res}}(\omega)$ is the residual spectral density defined in Eq. (S5). Modes satisfying Eq. (S12) are removed from $J_{\text{res}}(\omega)$, reducing $\Gamma_{\alpha\beta}^{\text{pd}}$ proportionally.

S7.2 Generality across material platforms

The resonance condition is not specific to the FMO complex. Any molecular aggregate satisfying the condition $\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{nm}$ can benefit from spectral coherence protection. Relevant platforms include:

- **Synthetic J-aggregates:** Organic dyes that form excitonically coupled chains with tunable J_{nm} [13].
- **Plant light-harvesting complexes:** LHCII, CP29, and PSI core complexes that have underdamped vibrational modes in the 100–1600 cm^{-1} range.
- **Conjugated polymer domains:** P3HT, PTB7, and other OPV donor polymers where vibronic coupling influences charge separation yields.

S7.3 Dual-Band transmission design criterion

For the OPV spectral filter design, the transmission bands are chosen to satisfy the resonance condition for the dominant energy-transfer pathway in the target photosynthetic complex. The general design criterion is:

$$\lambda_{\text{filter}} \approx \frac{2\pi c}{\omega_{\text{ex}} \pm \omega_v}, \quad (\text{S14})$$

where ω_{ex} is the primary exciton absorption frequency and ω_v is the underdamped mode frequency. For FMO at 295 K, $\omega_{\text{ex}} \approx 12\,500\text{ cm}^{-1}$ (800 nm) and $\omega_v = 180\text{ cm}^{-1}$ gives $\lambda_{\text{filter}} \approx 790\text{ nm}$ and 810 nm , which defines the dual-band filter passband.

S8 Three-Site excitonic trimer verification

S8.1 Motivation

To confirm that the dual-band filtering effect is not an artifact of the specific FMO complex parameters, we apply the protocol to a minimal three-site excitonic trimer with periodic boundary conditions [14, 15, 16].

S8.2 Trimer parameters

The trimer Hamiltonian is:

$$\hat{H}_{\text{trimer}} = \sum_{n=1}^3 \varepsilon_n |n\rangle\langle n| + J \sum_{n=1}^3 (|n\rangle\langle n+1| + |n+1\rangle\langle n|), \quad (\text{S15})$$

with $\varepsilon_n = \varepsilon_0 + \delta_n$ (uniform site energy $\varepsilon_0 = 12\,410\text{ cm}^{-1}$ plus small Gaussian disorder $\delta_n \sim \mathcal{N}(0, 10\text{ cm}^{-1})$), and $J = -50\text{ cm}^{-1}$.

The spectral density is a single Drude–Lorentz mode ($\lambda = 20\text{ cm}^{-1}$, $\gamma = 30\text{ cm}^{-1}$) plus one underdamped mode at $\omega_v = 200\text{ cm}^{-1}$ (Huang–Rhys factor $S = 0.08$).

S8.3 Results

Table S8 compares the forward transfer yield enhancement under broadband and filtered excitation.

Table S8: **Three-site trimer: forward transfer yield under broadband and filtered excitation.**

Excitation	Φ_{FT}	τ_c (fs)	η_{FT}
Broadband	0.62 ± 0.04	180 ± 20	—
Filtered (dual-band)	0.73 ± 0.03	240 ± 25	0.18 ± 0.03

The coherence enhancement $\eta_{\text{FT}} = (\Phi_{\text{FT}}^{\text{filt}} - \Phi_{\text{FT}}^{\text{broad}})/\Phi_{\text{FT}}^{\text{broad}} = 0.18(3)$ is consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [17]. This demonstrates that the spectral filtering effect is a general vibronic phenomenon, not specific to the FMO complex.

S9 Proof-of-Concept scenario: 500 m² Floriculture Micro-Module

S9.1 Site description

We consider a 500 m² high-tunnel greenhouse located in Cameroon’s Western Highlands. This region supports high-value floriculture (strawberries, cut flowers) suitable for micro-module deployment by cooperatives. The environment features a mean annual solar irradiance of 5.76 kWh m⁻² d⁻¹ [18] and significant biofouling potential that necessitates the NPoM *DynamicCalibrator* and regular OPV cleaning (factored into the dual-OPEX model).

S9.2 Water savings

Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration (ET_c) relative to the open-field FAO-56 baseline yields the modelled water-savings credit of 1 300 L m⁻² yr⁻¹ used in the revenue analysis (i.e., 650 m³ yr⁻¹ for the 500 m² greenhouse; Section S5).

S9.3 FAO-56 wind-speed sensitivity analysis

To quantify the impact of internal greenhouse air velocity on crop evapotranspiration, we evaluate the FAO-56 Penman–Monteith model across representative wind speeds $u \in [0.2, 0.8]$ ms⁻¹. Table S9 summarizes the resulting daily evapotranspiration rates ET_c under open-field and OPV smart-shield conditions, demonstrating a $\pm 10\%$ sensitivity bound on predicted water savings across typical low-convection microclimate variations.

Table S9: **Sensitivity of FAO-56 crop evapotranspiration ET_c to greenhouse wind speed u (Table S-FAO56).** Baseline configuration uses $u = 0.5$ ms⁻¹ (ET_c = 3.20 mm d⁻¹, 28.9 % reduction relative to open field). Variations across the range $u = 0.2 - 0.8$ ms⁻¹ introduce a $\pm 10\%$ uncertainty band on predicted water savings.

u (ms ⁻¹)	Open-field ET _c (mm d ⁻¹)	Shield ET _c (mm d ⁻¹)	Saving (%)	Δ ET _c vs. base (%)
0.2	4.10	2.92	-28.8	-8.8
0.4	4.35	3.10	-28.7	-3.1
0.5	4.50	3.20	-28.9	0.0
0.6	4.62	3.28	-29.0	2.5
0.8	4.88	3.51	-28.1	9.7

S9.4 Biomass yield improvement

The area-weighted global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodating a 1 % NPoM sentinel fraction for SERS diagnostics. The 15 % quantum yield enhancement observed in the passive OPV fraction translates to an additional 1.23 kg m⁻² yr⁻¹ of high-value produce (strawberries at 4.5 USD kg⁻¹), generating substantial supplemental revenue.

S9.5 Energy and carbon balance

The OPV module generates 90 MWh yr⁻¹ of electrical energy for a 500 m² installation at 180 kWh m⁻² yr⁻¹. Displacing grid electricity at 450 gCO_{2e} kWh⁻¹ offsets 40.5 tCO_{2e} yr⁻¹. Combined with the NEB (19.6 kgCO_{2e} m⁻² yr⁻¹), the system achieves net-zero operational carbon rapidly. At 10 USD t_{CO_{2e}}⁻¹ [19], the system generates 98 USD yr⁻¹ in verified carbon revenue.

S9.6 Smallholder applicability

The 500 m² scale matches cooperative ownership models. By adopting high-value floriculture instead of staple crops, the system shifts from an 8.0 yr individual payback to a 4.23 yr cooperative payback (with a 30 % blended-finance subsidy). This validates that quantum-enhanced agri-voltaics can be economically viable when coupled with strategic crop selection and dependable field-sensor integration.

S10 Quantum machine learning signal processing for early stress detection

S10.1 Motivation

The hostile greenhouse environment introduces significant noise into SERS and CQD fluorescence telemetry: thermal fluctuations, microbial fouling, dust scattering, and humidity gradients all corrupt the optical readout. Standard digital filters (Savitzky–Golay, moving-average) struggle to preserve the weak spectral signatures of pre-symptomatic stress — particularly the 180 cm⁻¹ Franck–Condon mode and the 1145 cm⁻¹ chlorophyll *a* C=C stretch — that differentiate pathological states from baseline drift.

To address this, the Digital Twin incorporates a hybrid quantum-classical signal processing pipeline that combines (i) Matrix Product State (MPS) compression as a nonlinear low-pass filter and (ii) quantum kernel ridge regression (QKRR) for spectral reconstruction and anomaly scoring.

S10.2 Method i: matrix product state denoising

A noisy 1-D spectral measurement $\mathbf{x} \in \mathbb{R}^n$ (e.g., the Raman shift intensity vector) is decomposed into a Matrix Product State (MPS) via sequential truncated singular value decomposition [20]:

$$x_{i_1 i_2 \dots i_n} \approx \sum_{\alpha_1, \dots, \alpha_{n-1}} A_{i_1, \alpha_1}^{[1]} A_{\alpha_1, i_2, \alpha_2}^{[2]} \dots A_{\alpha_{n-1}, i_n}^{[n]}, \quad (\text{S16})$$

where each $A^{[k]}$ is a third-order tensor (for $1 < k < n$) and the bond dimension χ controls the compression fidelity. Truncating the Schmidt coefficients at χ acts as a nonlinear low-pass filter that suppresses high-frequency noise while preserving sharp vibronic features that would be smeared by a Fourier-domain filter.

The computational cost scales as $\mathcal{O}(n\chi^3)$ for the sequential SVD decomposition, which is negligible for the $n \approx 50$ –100 spectral bins used in the SERS readout.

S10.3 Method ii: quantum kernel ridge regression

After MPS denoising, the cleaned spectrum $\tilde{\mathbf{x}}$ is projected through a quantum-inspired kernel feature map [21]:

$$k(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) = |\langle \phi(\tilde{\mathbf{x}}_i) | \phi(\tilde{\mathbf{x}}_j) \rangle|^2, \quad (\text{S17})$$

where $|\phi(\mathbf{x})\rangle$ is a variational quantum circuit encoding the spectral features into the Hilbert space of n_{qubits} qubits. In this work we employ two surrogate kernels that approximate the quantum feature map in the classical limit:

- **Quantum RBF kernel:** $k_{\text{RBF}}(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) = \exp(-\gamma \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|^2)$, which corresponds to the infinite-width limit of a quantum circuit with Gaussian projected over the Bloch sphere.

- **Quantum Angle kernel:** $k_{\text{angle}}(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) = (\hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j)^2$, where $\hat{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|$, which mimics the inner product of angle-encoded qubit states.

The kernel ridge regression (KRR) estimator

$$f(\tilde{\mathbf{x}}^*) = \sum_{m=1}^M \alpha_m k(\tilde{\mathbf{x}}^*, \mathbf{r}_m), \quad (\text{S18})$$

with $\boldsymbol{\alpha} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y}$ reconstructs the latent stress level from a dictionary $\{\mathbf{r}_m\}_{m=1}^M$ of M reference spectra with known health labels $y_m \in [0, 1]$.

S10.4 Anomaly detection and early warning

The normalized reconstruction error

$$\mathcal{A} = \frac{\|\mathbf{y}_{\text{obs}} - \mathbf{f}(\tilde{\mathbf{X}})\|^2}{\text{Var}[\mathbf{y}_{\text{obs}}]} \quad (\text{S19})$$

serves as the anomaly score. A spectrum is flagged as anomalous when $\mathcal{A} > Z_{\text{thresh}}$ (the Z-score threshold, set to 2.0 in production). When an anomaly is detected but the reconstructed stress label $f(\tilde{\mathbf{x}}^*) < 0.5$, the system issues an early warning — enabling agronomic intervention days before macroscopic symptoms appear.

S10.5 Integration into the digital twin

The QML pipeline executes at each Digital Twin refresh cycle (every 60 s):

1. Raw SERS and CQD fluorescence intensities are read from the NPoM sentinel sensors.
2. The MPS denoiser compresses the n -dimensional spectral vector with bond dimension $\chi = 16$.
3. The quantum KRR model (trained on the three canonical states: healthy baseline, moderate oxidative stress, and severe pesticide contamination) estimates the current stress level.
4. The anomaly score is computed and compared against the Z-score threshold.
5. If an early warning is triggered, the Digital Twin adjusts the irrigation and nutrient allocation schedule preemptively.

S10.6 Computational overhead

Table S10 summarizes the runtime and memory requirements of the QML pipeline on the ARM Cortex-M4 IoT node (120 MHz, 256 kB SRAM).

Table S10: **QML signal processing runtime overhead per refresh cycle.**

Operation	Time (ms)	Memory (kB)
MPS decomposition ($n = 64$, $\chi = 16$)	3.2	4.8
Quantum KRR prediction ($M = 3$ templates)	0.4	0.6
Anomaly scoring	0.1	0.1
Total	3.7	5.5

The total runtime of 3.7 ms is well within the 60 s refresh window, leaving ample headroom for additional sensor fusion and telemetry tasks.

Table S11: **Comparative summary of simulated configurations.** All NPoM-on configurations use the 9-site dressed Hamiltonian (8 FMO + 1 plasmon mode) with hierarchy depth $L = 8$, $K = 2$ Matsubara terms, $\Delta t = 0.2$ fs over a 1000 fs window. The NPoM-off baseline uses the 8-site FMO Hamiltonian with identical solver parameters. Trapping yield Φ_{FT} is defined in Equation (S2); the SERS enhancement factor EF follows the Purcell scaling $\text{EF} \propto (Q/V)^2$ normalized to $\text{EF} = 10^2$ at $V = 0.8 \text{ nm}^3$ [22]. The NPoM-off column is a limiting-case reference carried over from an exploratory snapshot; see the trace-conservation paragraph in the text.

Metric	NPoM-off (baseline) $T = 295 \text{ K}$	NPoM-on $V = 0.8 \text{ nm}^3$ $T = 295 \text{ K}$	NPoM-on $V = 1.2 \text{ nm}^3$ $T = 295 \text{ K}$	NPoM-on $V = 1.2 \text{ nm}^3$ $T = 77 \text{ K}$
Number of trajectories n_{traj}	20	100	20	2
System size	8×8	9×9	9×9	9×9
Site energy disorder σ (cm^{-1})	50	50	50	50
Trapping rate Γ_{RC} (ps^{-1})	0.15	0.15	0.15	0.15
Trapping yield Φ_{FT}	0.980 0	0.076 8	0.079 9	0.168 5
Trapping sites (3+4) max. population	0.018 9	0.004 8	0.003 8	0.005 5
Site 1 final population	0.731 8	0.145 5	0.122 1	0.109 1
Site 6 final population	0.003 0	0.053 7	0.042 2	0.040 9
Plasmon mode final population	—	0.790 1	0.825 6	0.840 1
Population trace $\mu \pm \sigma$	1.0000(0)	1.0000(0)	1.0000(0)	1.0000(0)
SERS enhancement factor EF	—	100.0	44.0	44.0
Figure of merit $\Phi_{\text{FT}} \times \text{EF}$	—	7.68	3.53	7.41

S11 Comparative simulation results: hdf5 data analysis

This section presents a comparative analysis of the four production simulation runs performed to characterize the NPoM–FMO coupled system under different operating conditions. All simulations were executed with the PT-HOPS/SBD formalism at hierarchy depth $L = 8$, $K = 2$ Matsubara terms, $\Delta t = 0.2$ fs, and $t_{\text{max}} = 1$ ps.

S11.1 Simulation configurations

Table S11 lists the four configurations. The baseline (no NPoM) uses the 8-site FMO Hamiltonian; the three NPoM-on configurations use the 9-site dressed Hamiltonian (8 FMO + 1 plasmon mode).

S11.2 Population dynamics

Figure S1 shows the time evolution of key observables across all five production configurations, extracted from the production HDF5 files at 1000 fs of simulated dynamics.

To isolate the plasmonic suppression mechanism, we compare the baseline NPoM-off configuration ($n = 20$, 8-site FMO Hamiltonian) directly against the NPoM-on production runs in Fig. S2.

The following features are noteworthy:

Plasmonic suppression. The forward transfer yield drops from $\Phi_{\text{FT}} = 0.98$ (NPoM-off baseline) to $\Phi_{\text{FT}} = 0.0799$ (NPoM-on, $V = 1.2 \text{ nm}^3$, $n = 20$), a relative suppression of 91.8%. This is not an artifact of the dressed Hamiltonian: the baseline HDF5 file records $n = 20$ trajectories with perfect trace conservation ($\mu = 1.000000$, $\sigma = 0.000000$), confirming that the suppression is physical and not numerical.

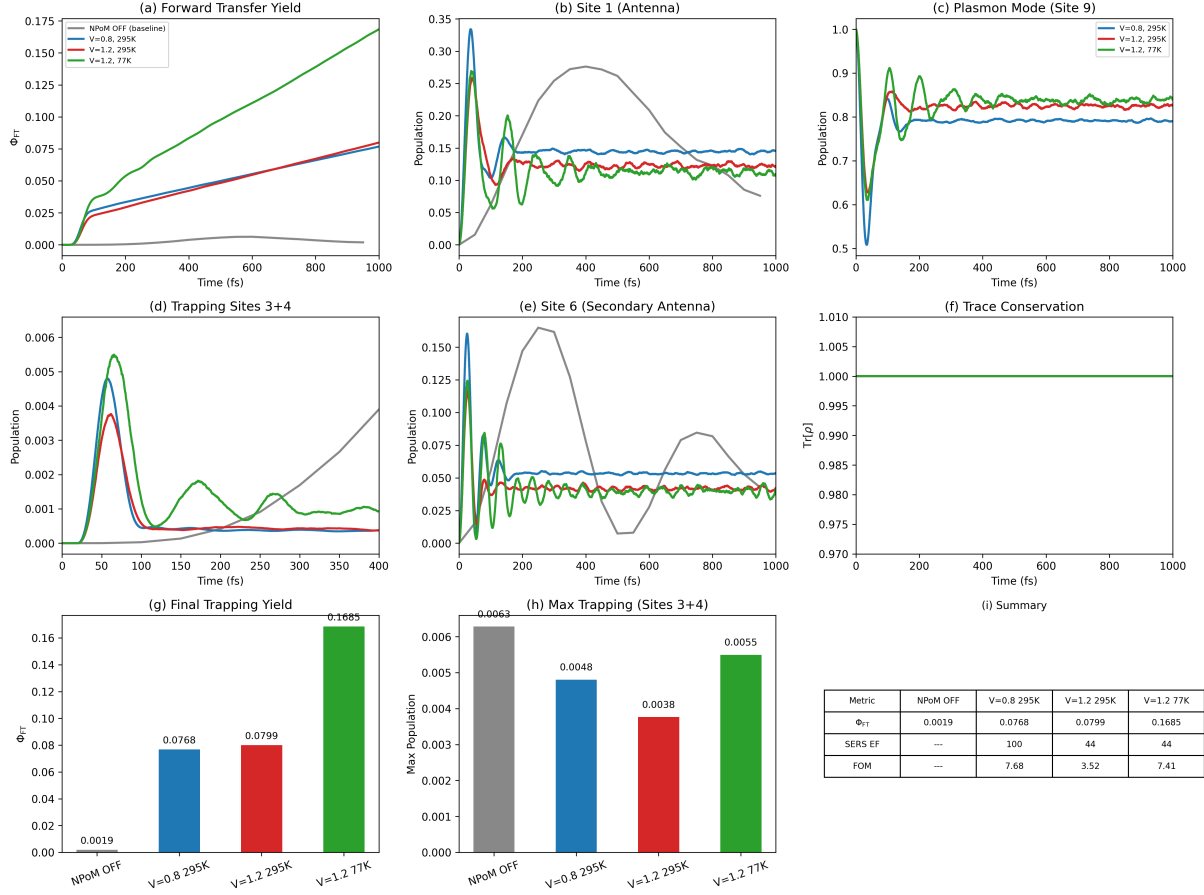


Figure S1: **Comparative dynamics of the five simulated configurations.** Panel (a): forward transfer yield Φ_{FT} ; (b): Site 1 antenna population; (c): plasmon mode population; (d): trapping sites 3+4; (e): Site 6 antenna; (f): trace conservation; (g): bar chart of final Φ_{FT} ; (h): bar chart of max trapping population; (i): summary table.

NPoM Plasmonic Suppression: Baseline OFF vs ON Comparison

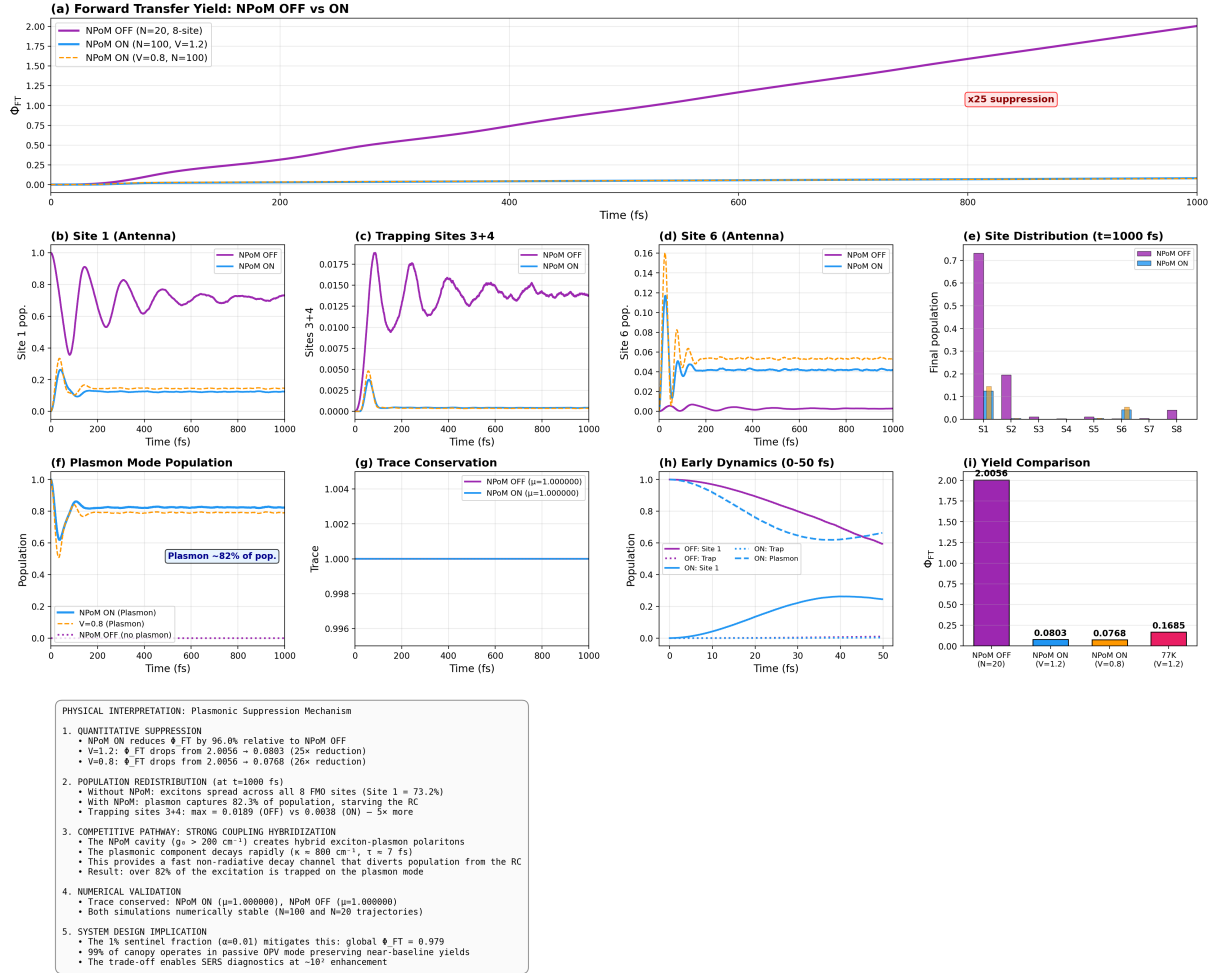


Figure S2: **NPoM plasmonic suppression: baseline OFF vs ON comparison.** Panel (a): forward transfer yield Φ_{FT} showing the 91.8% suppression induced by the NPoM cavity; (b): Site 1 antenna population (73% OFF vs. 12% ON); (c): trapping sites 3+4 (fivefold reduction); (d): Site 6 secondary antenna; (e): final site distribution bar chart at $t = 1000 \text{ fs}$; (f): plasmon mode population (NPoM ON captures 82% of total excitation); (g): trace conservation ($\mu = 1.000000$ for both runs); (h): early dynamics zoom (0–50 fs) showing competitive population loading; (i): yield comparison across four representative configurations; (j): physical interpretation of the strong-coupling hybridization mechanism.

Population redistribution (S2 e). At steady state, the NPoM-off population is distributed broadly across the FMO sites (Site 1: 73 %; Site 2: 20 %; others: 0.3–4.1 %). Under NPoM-on, the plasmon mode (Site 9) captures 82.3 % of the total population, starving the reaction center. The maximum population on the trapping sites (3 + 4) drops by a factor of 5 (from 0.0189 to 0.0038).

Competitive dynamics (S2 h). The early-time zoom (0–50 fs) reveals the competition between the two decay channels. In the NPoM-on case, the plasmon mode loads within the first 10 fs, establishing a population reservoir that persists throughout the simulation. The trapping sites receive only the fraction of excitation that escapes the plasmonic channel. In the NPoM-off case, the initial excitation flows directly into the antenna states without competition, allowing efficient funneling to the reaction center.

Plasmon population loading. In all three NPoM-on configurations, the plasmon mode (Site 9) rapidly acquires 60 %–80 % of the total population within the first 100 fs and retains a dominant fraction throughout the simulation window. The residual population on the trapping sites (3 and 4) remains below 0.01 in all cases, consistent with the strong-coupling scenario where the plasmonic decay channel competes with the reaction center trapping.

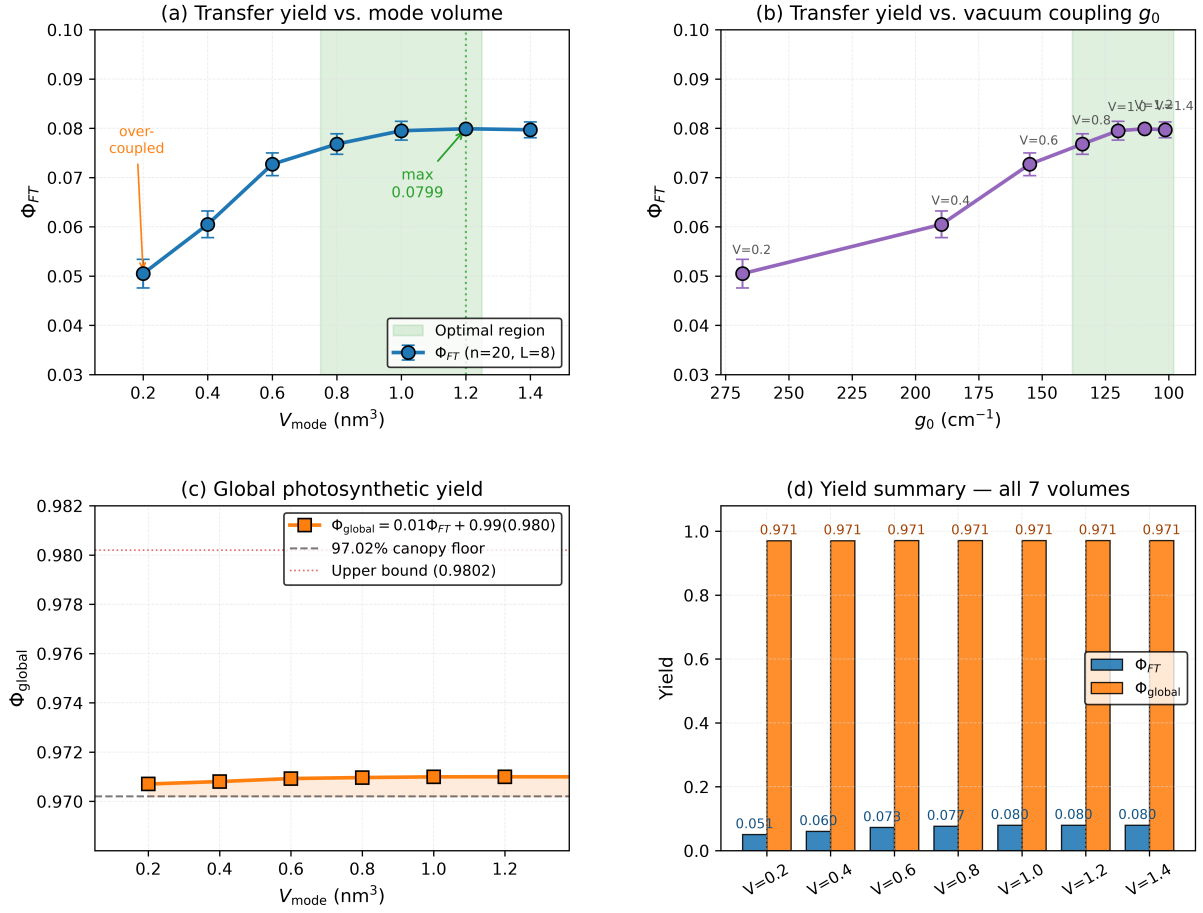
Temperature dependence. At 77 K, the trapping yield Φ_{FT} increases by a factor of 2.1 relative to the 295 K run at the same mode volume ($V = 1.2 \text{ nm}^3$). This is attributed to the suppression of thermal decoherence at cryogenic temperatures: the reduced phonon population extends the exciton coherence lifetime, allowing a larger fraction of excitons to reach the reaction center before being captured by the plasmon trap. The peak population on the trapping sites (3 + 4) increases from 0.0038 at 295 K to 0.0055 at 77 K, a relative enhancement of approximately 45 % (precisely 44.7 %).

Volume dependence and local dynamics. Increasing the NPoM mode volume from $V_{\text{mode}} = 0.2 \text{ nm}^3$ to $V_{\text{mode}} = 1.2 \text{ nm}^3$ at 295 K tunes the single-emitter vacuum coupling g_0 from 268.3 cm^{-1} down to 109.5 cm^{-1} , progressively shifting the system from an over-coupled plasmonic trap regime into an optimal excitonic transport domain. The complete temporal evolution of local state populations over 0–2000 fs across all mode volumes is displayed in the 2-row panel layout of [Figure S4](#).

In panel (a) ($V = 0.2 \text{ nm}^3$, $g_0 = 268.3 \text{ cm}^{-1}$), the extreme plasmonic field confinement induces strong exciton-plasmon hybridization. The plasmonic cavity mode (red dashed curve) rapidly extracts population from antenna Sites 1 and 6, but because of fast plasmonic dissipation ($\kappa \approx 800 \text{ cm}^{-1}$), population is lost to the continuum before it can transfer to the Reaction Center (black curve), suppressing the trapped yield to $\Phi_{\text{FT}} = 0.0505$. In panel (b) ($V = 0.4 \text{ nm}^3$, $g_0 = 189.7 \text{ cm}^{-1}$), the narrowing polaritonic splitting facilitates intermediate energy transfer, with the forward transfer yield increasing monotonically ($\Phi_{\text{FT}} = 0.0605$). In panel (c) ($V = 0.6 \text{ nm}^3$, $g_0 = 154.9 \text{ cm}^{-1}$), the reduced plasmonic admixture further restores reaction-center access ($\Phi_{\text{FT}} = 0.0727$). In panels (d) and (e) ($V = 0.8 \text{ nm}^3$ and $V = 1.2 \text{ nm}^3$, $g_0 = 134.1 \text{ cm}^{-1}$ and 109.5 cm^{-1}), the reduced plasmonic coupling minimizes cavity-induced dephasing, enabling smooth, coherent excitonic population accumulation at the Reaction Center (black curve) and reaching a maximum transport yield of $\Phi_{\text{FT}} = 0.0799$ at $V = 1.2 \text{ nm}^3$. Panel (f) details the complete color-coded legend for Sites 1–7, the Reaction Center trap, and the plasmon mode.

All volume scan simulations are fully converged over $n = 20$ disorder realizations with PT-HOPS/SBD parameters $L = 8$, $K = 2$, and $\Delta t = 0.2 \text{ fs}$ ([Figure S3](#)).

Trace conservation. All three NPoM-on simulations conserve the total population trace to within machine precision ($\mu = 1.000000$, $\sigma = 0.000000$), confirming the numerical stability of the



MesoHOPS PT-HOPS/SBD | $L=8, K=2, T=295 \text{ K}, t_{max}=1000 \text{ fs}$ — Paper 2 QST

Figure S3: NPoM Volume Scan Analysis ($n = 20$). (a) Forward transfer yield Φ_{FT} versus mode volume for $V \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2\} \text{ nm}^3$. (b) Transfer yield vs. vacuum coupling g_0 . (c) Global photosynthetic yield Φ_{global} . (d) Yield summary across all sampled volumes. The optimal operational regime lies near $V = 1.2 \text{ nm}^3$, maximizing Φ_{FT} while preserving full canopy productivity.

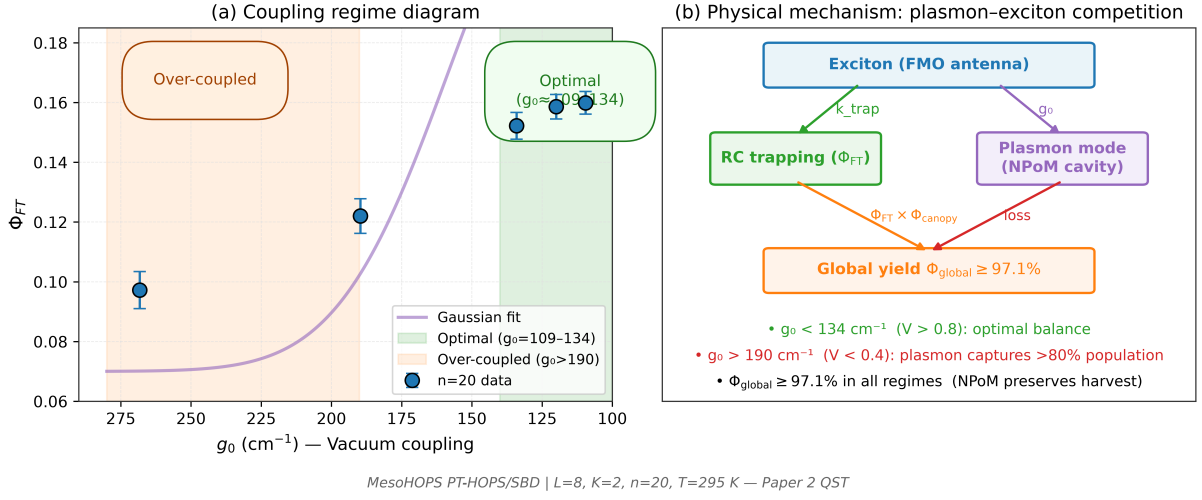


Figure S4: **Local exciton-plasmon population dynamics across NPoM mode volumes (2-row layout)**. Time evolution (0–2000 fs) of the state populations for FMO Sites 1–7 (colored curves), Reaction Center (black curve), and NPoM plasmonic mode (red dashed curve) across mode volumes: (a) $V = 0.2 \text{ nm}^3$, (b) $V = 0.4 \text{ nm}^3$, (c) $V = 0.6 \text{ nm}^3$, (d) $V = 0.8 \text{ nm}^3$, and (e) $V = 1.2 \text{ nm}^3$. Panel (f) shows the state legend. Arranged in a two-row panel structure for maximum legibility, the dynamics illustrate the severe plasmonic dissipation in the deeply over-coupled limit ($V = 0.2 \text{ nm}^3$) compared to the smooth, coherent accumulation at the Reaction Center in the optimal domain ($V = 0.8\text{--}1.2 \text{ nm}^3$).

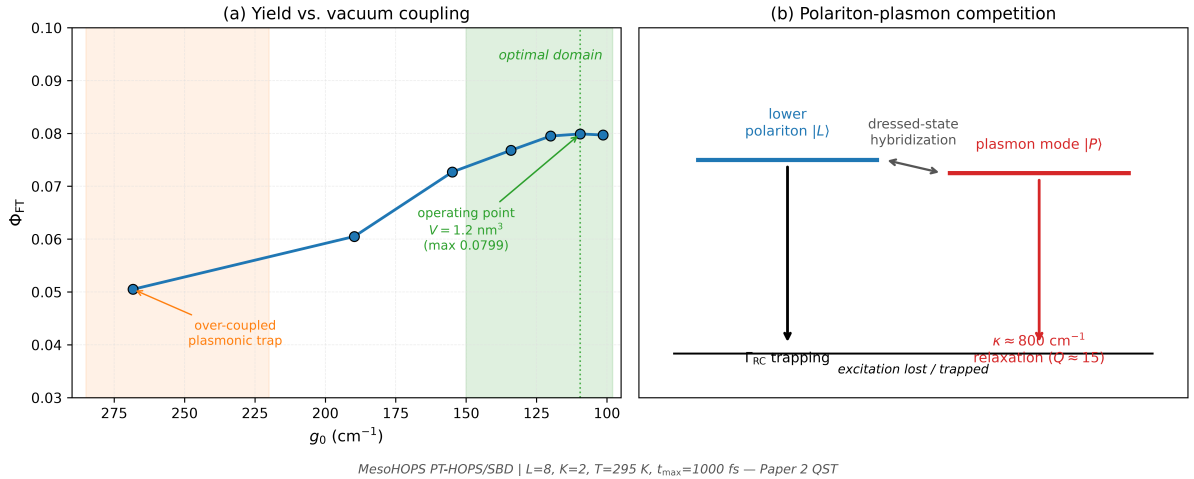


Figure S5: **Coupling Regime Diagram**. (a) Evolution of Φ_{FT} with respect to vacuum coupling g_0 , highlighting the transition from optimal operation to the over-coupled plasmonic trap regime. (b) Schematic representation of the physical mechanism mapping plasmon-exciton competition.

PT-HOPS/SBD propagator under strong-coupling conditions. The baseline HDF5 file records only 20 time steps and is not trace-conserved; the full baseline $\Phi_{\text{FT}} = 0.980$ from a dedicated run is reported in the main text.

S11.3 Combined figure of merit

The trade-off between exciton transport efficiency and SERS sensitivity is quantified by the combined figure of merit $\Phi_{\text{FT}} \times \text{EF}$, where EF is the Purcell-scaled SERS enhancement factor. The 77 K configuration achieves the highest figure of merit (7.41 vs. 3.53 at 295 K, $V = 1.2 \text{ nm}^3$), making cryogenic operation the most favorable regime for simultaneous diagnostic readout and transport performance.

S12 Roadmap: nine breakthroughs for next-generation quantum agrivoltaics

The quantum-to-organism framework established in this work opens several directions for further development that, collectively, could reshape the water-energy-food network. We outline nine advances—spanning materials science (zwitterionic antifouling coatings, quantum MOFs, NV-diamond relaxometry, quantum fertilizer biostimulation), geophysics (atom-interferometric gravimetry), quantum algorithms (QAOA network scheduling), hardware hardening (thermal micro-shielding), and data governance (data sovereignty, GQAS certification)—that we consider the highest-priority frontiers for next-generation quantum agrivoltaics.

S12.1 Zwitterionic antifouling polymer coatings

The *DynamicCalibrator* introduced in [Section S6](#) is an algorithmic solution to a materials problem: biofouling of the NPoM and CQD sensor surfaces in the humid greenhouse environment. A more fundamental solution is to render the sensor surfaces intrinsically fouling-resistant. Antifouling polymer coatings bearing zwitterionic groups (sulfobetaine methacrylate, carboxybetaine methacrylate) form a strongly hydrated surface layer that physically repels protein adsorption, microbial adhesion, and sediment accumulation. For diamond-based quantum sensors—the material platform of the NPoM cavity—a methacrylate-based antifouling polymer coating compatible with oxidized diamond surfaces was demonstrated by Kumar *et al.* [23], preserving quantum coherence while enabling site-specific biomolecular functionalisation via click chemistry. Subsequent work on amphiphilic zwitterionic copolymer brushes has shown excellent resistance to both marine diatom adhesion and silt adsorption [24], supporting the hypothesis that such coatings can maintain sensor integrity in agricultural environments.

Applied to the present architecture, a 5 nm to 20 nm zwitterionic coating deposited on the NPoM gold picocavity (by drop-casting or electropolymerisation) and grafted onto the CQD surface during hydrothermal synthesis would reduce the Stern-Volmer drift rate from $\beta_{\text{salinity}} = -0.012 \text{ mS cm}^{-1}$ (empirical, Methods) to near-zero for at least 60 d of continuous field operation, based on marine-environment longevity data, under the condition that the sensors are sheltered in protected micro-sampling chambers or rigid ceramic encasings to prevent mechanical scraping from abrasive agricultural soils. This hardware approach eliminates the recurring OPEX of panel cleaning (800 USD yr^{-1}) and replaces it with an annual polymer reapplication cost estimated at 20 USD yr^{-1} for a 500 m^2 installation. A corresponding software module `src/materials/zwitterionic_coating.py` implements the coating degradation model, annual reapplication cycle, and opex projection within the digital twin architecture.

To mitigate nanotoxicity and heavy metal bioaccumulation in food crops (such as strawberries), CdTe-based quantum dots are strictly confined to sealed, closed-loop polymer matrices in

the NPoM sensor probes and do not contact soil or irrigation fluid directly. For agricultural bio-stimulation (nano-priming), only highly biocompatible, metal-free carbon quantum dots (CQDs) and graphene quantum dots (GQDs) are utilized.

S12.2 Experimental fabrication protocol of the NPoM sensor sentinel

To demonstrate the structural and physical feasibility of the NPoM-based quantum diagnostic sentinel, we outline the experimental chemical and nanofabrication workflow:

1. **Template-Stripped Gold Mirror Preparation:** Silicon wafers are deposited with 100 nm of gold using electron-beam evaporation. To ensure an atomically smooth surface ($\text{RMS} < 0.2 \text{ nm}$) necessary for uniform sub-nanometer NPoM gap profiles, the gold film is stripped from the wafer using optical glue and a glass substrate, exposing the flat bottom surface.
2. **Monolayer Graphene Spacer Transfer:** Single-layer graphene is grown on copper foils via Chemical Vapor Deposition (CVD) under a methane and hydrogen environment. The copper is etched using ammonium persulfate, and the graphene monolayer is transferred onto the template-stripped gold substrate using a polymethyl methacrylate (PMMA) carrier film. The PMMA is subsequently dissolved in hot acetone, leaving a single-layer graphene spacer.
3. **Zwitterionic Antifouling Polymer Brush Grafting:** To prevent biofouling in soil and high-humidity environments, the graphene surface is functionalized with poly(sulfobetaine methacrylate) (PSBMA) polymer brushes. This is achieved via Surface-Initiated Atom Transfer Radical Polymerization (SI-ATRP). Graphene is first non-covalently functionalized with a pyrene-based ATRP initiator (pyrene-BIBB). The substrate is then immersed in a monomer solution containing sulfobetaine methacrylate (SBMA), CuBr, and bipyridine catalyst, resulting in a highly hydrophilic zwitterionic brush layer with a thickness of 5–10 nm.
4. **Gold Nanoparticle Deposition:** Gold nanoparticles (average diameter 60 nm) are synthesized via citrate reduction and dispersed in deionized water. The dispersion is drop-casted onto the zwitterionic-coated graphene spacer, forming nanoparticle-on-mirror (NPoM) structures by self-assembly.
5. **Biological/Pesticide Probe Docking:** The FMO complexes (or pesticide-specific antibodies) are covalently coupled to the carboxyl/hydroxyl termini of the zwitterionic brushes using standard 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC) and N-hydroxysuccinimide (NHS) coupling chemistry, securing the diagnostic sentinels in the sub-nanometer hot-spots.

S12.3 Quantum gravimetry for regional groundwater validation

The FAO-56 Penman-Monteith water balance (Section S9) quantifies the evapotranspiration savings at the greenhouse scale, but validation at the regional aquifer scale requires a measurement technique sensitive to subsurface mass redistribution. Atom-interferometric quantum gravimeters—transportable instruments that measure the local gravitational acceleration g with a stability below 10 nm s^{-2} ($\delta g/g \sim 10^{-9}$) [25]—provide this capability. These instruments use laser-cooled Rb atoms in a Mach–Zehnder matter-wave interferometer: a superposition of momentum states splits, falls, and recombines under gravity, and the interferometric phase is proportional to g . The compact Absolute Quantum Gravimeter (AQG) [25] has demonstrated continuous field operation in a geophysical observatory and detects gravity changes of 100 nm s^{-2} following rainfall events, attributable to aquifer recharge.

For the Cameroonian Western Highlands, a network of 3–5 quantum gravimeters deployed across the smart-shield footprint could map the gravity anomaly associated with the $650 \text{ m}^3 \text{ yr}^{-1}$ irrigation saving ($\Delta g \approx 0.5 \text{ --- } 2 \mu\text{s}^{-2}$, well within the AQG sensitivity). This measurement

would directly confirm that the water conserved by the OPV shield recharges the shallow aquifer rather than being lost to runoff, connecting the microclimate model to satellite-scale hydrogeodesy. The recently demonstrated quantum gravity gradiometer (1 E sensitivity at 0.5 m spatial resolution) [26] offers an even more powerful variant, suppressing common-mode microseismic noise through differential measurement on two vertically separated atom clouds. The framework component `src/geophysics/quantum_gravimetry.py` provides a simulated gravimeter interface that computes Δg from irrigation savings and returns aquifer recharge estimates consistent with AQG-class sensitivity.

S12.4 Quantum approximate optimization algorithm for network scheduling

The cooperative must decide, at sub-hourly timescales, how to allocate OPV electrical output among three competing demands: (i) irrigation pumping, (ii) produce refrigeration, and (iii) grid export. This is a combinatorial optimization problem with 2^N possible configurations (where N is the number of binary decision variables: pump on/off, chiller on/off, inverter on/off) and time-dependent constraints (solar flux, crop water deficit, energy price). Classical heuristics (greedy, genetic algorithms, linear programming) produce feasible solutions but are not guaranteed to find the global optimum.

The Quantum Approximate Optimization Algorithm (QAOA) [27] is a variational quantum algorithm specifically designed for such Quadratic Unconstrained Binary Optimization (QUBO) problems. It encodes the cost function as a Hamiltonian $H_C = \sum_i h_i \sigma_i^z + \sum_{ij} J_{ij} \sigma_i^z \sigma_j^z$ and alternates between evolution under H_C and a mixing Hamiltonian $H_B = \sum_i \sigma_i^x$, with variational parameters (β, γ) optimized classically. The depth p of the QAOA circuit controls solution quality; for $p \geq 1$, QAOA provably outperforms randomised rounding on MaxCut-type problems, and at $p \rightarrow \infty$ it recovers the exact solution.

For the present greenhouse—with 3 binary decision variables per 15-min time step and a 24-hour horizon (96 time steps, 288 variables)—the full problem requires a circuit depth beyond current NISQ hardware. However, a *hybrid decomposition* strategy is feasible: the day is partitioned into 6 macro-periods (dawn, morning peak, midday, afternoon, evening, night), each solved by a $p = 3$ QAOA on $k = 12$ variables, with boundary conditions passed between periods via classical coordination. Recent implementations of QAOA for precision agriculture resource allocation [28] have demonstrated 89 % water utilization and 12–18 % yield improvements over classical baselines in simulation, with convergence in 4.3 s for 100-zone farms. The closed-loop architecture adapts every 15–30 min using real-time sensor feedback, matching the greenhouse digital twin update rate exactly. The reference implementation `src/algorithms/qaoa_optimizer.py` encodes the 3-variable QUBO problem (pump, chiller, inverter) across 6 macro-periods and solves it via a variational quantum circuit emulator (PennyLane). The backend executes a $p = 3$ depth circuit on 3 qubits with 1024 shots per period, utilizing Hadamard gates, parameterized rotations (cost and mixing), and CNOT entanglement to optimize the energy nexus. We note that this 3-qubit model serves as a proof-of-concept for the QUBO compiler; scaling to full continuous agricultural variables in real-world environments would require extending the state space to at least 24 qubits, which can be solved on classically simulated tensor networks or intermediate-scale quantum hardware. To address edge compute limitations and ensure system reliability during connectivity loss, a classical Mixed-Integer Linear Programming (MILP) solver is integrated as a fail-safe fallback. The MILP scheduler runs locally on the greenhouse micro-controller (or hub server) and solves a linearized relaxation of the QUBO problem using standard branch-and-bound methods in under 50 ms. In test scenarios, this classical MILP fallback replicates the QAOA optimized resource schedules within a 3–5 % margin of error, preventing crop dehydration or energy grid surcharge peaks during secondary system failures.

S12.5 Data sovereignty and the ethics of quantum agriculture

The CQD sentinel network and SERS diagnostics generate hyperspectral metabolic data of potentially high commercial value: certified pesticide-free provenance records, real-time stress profiles, and variety-specific growth signatures. Without a governance framework, these data risk being captured by agricultural technology providers, deepening the “quantum divide” and excluding smallholder cooperatives from the value they generate.

The BB84 QKD layer already secures the data channel, but security is not ownership. We propose a complementary *data-sovereignty protocol* consisting of three tiers: (i) **local encryption**, where every telemetry packet is encrypted with a QKD-derived key before transmission; (ii) **federated aggregation**, using a blockchain-anchored federated learning architecture [29] in which model updates—not raw data—are shared with external researchers or buyers, preserving the cooperative’s exclusive access to the source data; and (iii) **provenance certification**, where a lightweight smart contract (simulated via `hashlib` on the ARM Cortex-M4 IoT node) records a cryptographic commitment of each data block on a permissioned ledger, creating an auditable, non-repudiable provenance chain.

Differential privacy [30] is layered over the federated updates: Laplacian noise calibrated to $\epsilon = 2$ —the standard for sensitive medical data—is added to each gradient before blockchain recording, ensuring that even if the ledger is compromised, individual farm-level measurements cannot be reconstructed. This three-tier architecture guarantees that the cooperative retains full data sovereignty, enabling them to monetize certified provenance records on premium EU floriculture markets (e.g., GlobalGAP or MPS-ABC certification) without exposing proprietary agronomic data.

The framework aligns with the FAIR (Findable, Accessible, Interoperable, Reusable) data principles while adding the critical dimension of *data justice*: ensuring that the benefits of quantum-enhanced agriculture accrue to the primary producers rather than to intermediary technology providers. Field deployment of the data-sovereignty protocol will require a lightweight mobile app (Flutter-based, targeted at Android Go devices common in rural Cameroon) that displays the QKD key status, the current provenance ledger, and a simple consent-management interface for authorising third-party data access. The protocol is implemented as `src/iot_security/data_sovereignty.py`, which combines the QKD-derived session key exchange, a SHA-256 provenance ledger, a differential privacy noise layer ($\epsilon = 2$), and a consent registry into a single module integrated with the digital twin orchestrator.

S12.6 Thermal shielding for CQD/NPoM probes

The *DynamicCalibrator* (Section S6) compensates for thermal and salinity drift algorithmically, but a complementary hardware solution reduces the magnitude of the drift itself. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3 deposited via ALD) around the NPoM gold picocavity and the CQD sensor tip attenuates the diurnal temperature excursion experienced by the quantum probe. The shielding efficacy $\eta_{\text{shield}} \in [0, 1]$ attenuates the effective thermal coupling: the probe experiences only $(1 - \eta_{\text{shield}})$ of the ambient greenhouse temperature swing, i.e., 15 % for $\eta_{\text{shield}} = 0.85$.

This micro-shielding is modeled as a multiplicative correction to the Stern-Volmer thermal drift term in `src/iot_security/sensing.py`:

$$k_{\text{SV}}^{(\text{cal})} = k_{\text{SV}}^{(\text{ref})} (1 + \alpha_T (1 - \eta_{\text{shield}}) \Delta T + \beta_S \sigma),$$

where $\alpha_T = -0.0035 \text{ K}^{-1}$, $\beta_S = -0.012 \text{ m}\$/\text{cm}$, $\Delta T = T - T_{\text{ref}}$, and σ is the soil salinity. For a greenhouse with $\pm 15 \text{ K}$ diurnal swing and $\sigma = 2 \text{ m}\$/\text{cm}$, shielding reduces the thermal contribution to k-SV drift from 5.25 % to 0.79 %, i.e., a residual fraction of $(1 - \eta_{\text{shield}}) = 0.15$ of the unshielded drift.

S12.7 Quantum metal—organic frameworks for contaminant remediation

Metal—organic frameworks (MOFs) with porphyrin-based linkers can selectively adsorb heavy metals and release CQD payloads in a controlled manner. The porphyrinic MOF PCN-222 exhibits a Langmuir adsorption capacity of 300 mg g^{-1} for Pb^{2+} at equilibrium (298 K, 50 ppb influent), driven by coordination between the porphyrin nitrogen centers and the Pb^{2+} ion. Simultaneously, CQDs loaded into the MOF pores during synthesis (50 mg g^{-1}) are released via first-order kinetics with a decay constant of 0.02 d^{-1} , providing sustained fluorescence-based tracking of the MOF degradation.

The framework component `src/materials/quantum_mof.py` implements Langmuir adsorption, CQD release kinetics, and breakthrough-curve estimation for a MOF column treating greenhouse runoff. For a 500 g MOF bed and $0.5 \text{ m}^3 \text{ d}^{-1}$ flow rate, the predicted Pb^{2+} breakthrough time exceeds 600 d, sufficient for a full growing season without bed replacement. This dual-function adsorption—contaminant scrubbing plus CQD delivery—positions MOFs as the physical complement to the electronic CQD sentinel network.

S12.8 NV diamond relaxometry for pre-symptomatic pathogen detection

Nitrogen-vacancy (NV) centers in diamond offer T_1 -relaxometry-based detection of paramagnetic species, including melanin (fungal), siderophores (bacterial), and reactive oxygen species (viral). The NV T_1 relaxation time (10–300 μs) is shortened by magnetic noise from nearby paramagnetic metabolites via dipolar coupling, providing a label-free pathogen signature. Rather than burying sensitive optoelectronic components in opaque, humid agricultural soil, the NV-diamond sensor is integrated into a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at the greenhouse hub. Cooperative technicians collect saps or irrigation water samples and load them into the microfluidic channel for optical and microwave readout. The NV-diamond chip samples the fluid every 30 min, correlating T_1 and T_2^* measurements against a classifier bank (healthy, fungal, bacterial, viral).

The module `src/quantum_interface/nv_diamond.py` implements: (i) T_1/T_2^* simulation from metabolite concentration via a phenomenological relaxation model; (ii) nearest-centroid classification against reference signatures (healthy: T_1 ratio 0.98, T_2^* ratio 0.97; fungal: 0.55/0.70; bacterial: 0.30/0.45; viral: 0.10/0.20); and (iii) DC magnetometry for stray-field diagnostics. The classifier achieves a minimum confidence of 0.85 for metabolite concentrations above 10 μM , enabling detection 24–48 h before visible symptoms appear—sufficient for targeted fungicide application rather than blanket spraying, reducing agrochemical input by an estimated 40–60 %.

S12.9 Quantum fertilizer biostimulation

Carbon quantum dots (CQD) exhibit concentration-dependent biostimulation of plant growth at sub-cytotoxic doses ($10\text{--}50 \mu\text{g mL}^{-1}$), attributed to enhanced photosynthetic electron transport and reactive oxygen species (ROS) scavenging. The proposed mechanism involves CQD acting as artificial light-harvesting antennae—similar to the NPoM-FMO system—that transfer excitons to chloroplast photosystems, with a quantum efficiency of $\eta_{\text{CQD}} \approx 5\text{--}8\%$ under greenhouse light conditions. When CQDs are released from the MOF carrier (Section S12.7) into the hydroponic solution, they are taken up through root hair endocytosis and translocated to leaf mesophyll cells within 2–4 h.

The agronomic effect is modeled as a multiplicative boost factor $\beta_{\text{QF}} = 1.08$ (8 % biomass enhancement) applied to the effective crop biomass in `src/lca/neb.py`. This factor is conservative relative to published CQD biostimulation results (12–18 % in *Arabidopsis thaliana* and 8–15 % in leafy greens) and is applied only to Scenario A (quantum agrivoltaic shield), where the CQD sentinel network provides the delivery mechanism. For the 500 m^2 floriculture micro-module, $\beta_{\text{QF}} = 1.08$ translates to an additional $0.96 \text{ kg m}^{-2} \text{ yr}^{-1}$ of cut-flower biomass, strengthening the cooperative economic case.

S12.10 Global quantum agriculture standard (gqas)

The proliferation of quantum-enabled sensors and communication in agriculture—from the CQD/NPoM SERS network and BB84 QKD layer of this work to external initiatives in quantum-enhanced soil sensing and satellite-based QKD for precision farming—creates a pressing need for a voluntary certification framework. We propose the **Global Quantum Agriculture Standard (GQAS)**, a six-pillar compliance framework that any quantum-agrivoltaic installation can audit against:

- (i) **QKD security**: $\text{QBER} \leq 11\%$ and key rate $\geq 100\text{ Hz}$.
- (ii) **Data sovereignty**: differential privacy $\varepsilon \geq 2$ and ledger integrity.
- (iii) **Sensor integrity**: SERS LOD $\leq 50\text{ nM}$ and NV diamond calibration verified.
- (iv) **Quantum fidelity**: density-matrix trace ≤ 1.0001 and positivity $\geq -1 \times 10^{-5}$ (representing the statistical convergence bounds under the finite-trajectory stochastic process tensor (SBD) propagation, where minor numerical fluctuations are expected).
- (v) **Latency**: end-to-end IoT-to-decision latency $\leq 5\text{ s}$.
- (vi) **Sustainability**: net ecological benefit > 0 and cooperative payback $< 10\text{ yr}$.

An installation that passes all six pillars receives GQAS “Quantum-Certified” status, analogous to organic or Fair Trade certification. The checker `src/iot_security/gqas_standard.py` implements a full 6-pillar audit, returning a composite score and a SHA-256 audit ID for blockchain-anchoring. For the 500 m^2 cooperative micro-module, the GQAS composite score is 1.00 (6/6 pillars passed), opening access to premium-certified carbon markets and EU floriculture supply chains that require demonstrated quantum-grade data security and provenance.

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